Causal Machine Learning assignment

Thomas Sertijn, Bart Smets, Ilja Van Bever

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Introduction

In this project the effect of eligibility for enrolling in a 401(k) plan on net financial assets is investigated.

1a) Effect estimation with Causal Forests

In the first phase causal forests were used to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets, in function of age, income, family size, years of education, a married indicator, a two-earner status indicator, a defined benefit pension status indicator, an IRA (individual retirement account) participation indicator, and a home ownership indicator.

The causal forest analysis reveals a causal effect of eligibility for enrollment in a 401(k) plan on accumulated net financial assets. The estimated Average Treatment Effect (ATE) equals 7799.81, indicating that individuals who are eligible for a 401(k) plan accumulate, on average, approximately $7799.81 more in net financial assets than non-eligible individuals. The estimated standard error is $7799.81, resulting in a 95% confidence interval of [$5477.19, 1.012243×10^4]. Because the confidence interval is positive and larger than 0, we conclude that being eligible for the 401(k) has a significant and positive effect on the net financial assets of a person.

In addition to the average treatment effect, the Conditional Average Treatment Effect (CATE) was examined to investigate heterogeneity in treatment effects across individuals with different covariate profiles. Figure 2 presents a histogram of the predicted CATE values. Most individuals have positive CATE values, although negative values are also observed. Thus, this suggests that for most covariate profiles, the treatment effect was expected to be positive, yet not for all profiles. Furthermore, the estimated CATE was plotted against income (Figure 2), revealing a clear upward trend: the effect of 401(k) eligibility appears to increase with income. Thus, becoming eligible does have a positive impact on financial assets, but more strongly so for richer individuals.

To better characterize this relationship, both a linear regression model and a smoothing spline regression (with $\lambda = 1$) were fitted. The corresponding trends are displayed in Figures 2 and 2. While the estimated curves differ somewhat at higher income levels, this part of the income distribution contains relatively few observations, implying considerably greater uncertainty in this region.

To assess the stability of the inverse probability weights, the distribution of the estimated propensity scores was also investigated using a histogram (Figure 2). 0% of the scores were smaller than 0.01 or larger than 0.99, and only 0.02% of the scores were smaller than 0.05 or larger than 0.95. Overall, the propensity scores do not exhibit problematic extremes, so truncation of propensity scores is not needed.

To estimate the propensity scores and outcomes, 5-fold cross-fitting was applied. The modeling was performed using Regression Forests, as described in the documentation of the GRF package (https://grf-labs.github.io /grf/reference/causal_forest.html).

1b) Possible problems when the analysis additionally conditions on other variables

It was previously stated that eligibility for a 401(k) could be taken as exchangeable conditional on income. However, conditioning on additional variables can have important implications depending on their role in the causal structure.

Role of Additional Variables in a Causal Diagram

Including variables in an analysis can have different effects depending on their role in the causal diagram. We illustrate this with figures below.

- *Collider*
  In the case of a collider *(Eligibility → Collider ← Outcome)*, conditioning on this variable introduces collider bias. Normally, this path would be blocked. However, conditioning on the collider opens this path and creates a spurious association between eligibility and net financial assets, which would bias our treatment effect. *IRA* could for example act as a *collider*, since individuals who are ineligible for a 401(k) may substitute toward an IRA as an alternative retirement savings vehicle, while wealthier individuals with higher net financial assets are also more likely to participate in an IRA.

- *Mediator*
  In the case of a mediator *(Eligibility → Mediator → Outcome)*, part of the effect of eligibility is captured through an indirect effect (via the mediator). Conditioning on a mediator is therefore not incorrect, but it implies that the estimated effect is no longer the total effect.

- *Common Cause*
  In the case of a common cause *(Eligibility ← Common Cause → Outcome)*, it is necessary to condition on this variable in the analysis. If a common cause is omitted, a backdoor pathway remains open, making the estimated effect of eligibility biased. This is because eligible and non-eligible individuals would then differ in characteristics, which also affect the outcome. We note that conditional exchangeability given only income is a strong assumption and this does not hold if there exist common causes of eligibility and the outcome. *Age* could for example potentially be a common cause. Older individuals are generally more likely to be eligible for a 401(k) program (stable full-time jobs, longer job tenure), while older people have in general more financial assets. Another example of a possible common cause could be *Years of Education*, because it can influence both *Eligibility* and *financial assets* in a positive way.

- *Variable Affecting Only Eligibility*
  If a variable only affects eligibility, including it in the analysis does not introduce bias.

- *Variable Affecting Only the Outcome*
  If a variable only affects the outcome and not eligibility, it can be included in the analysis.
  Including such variables may improve the efficiency of the estimation.

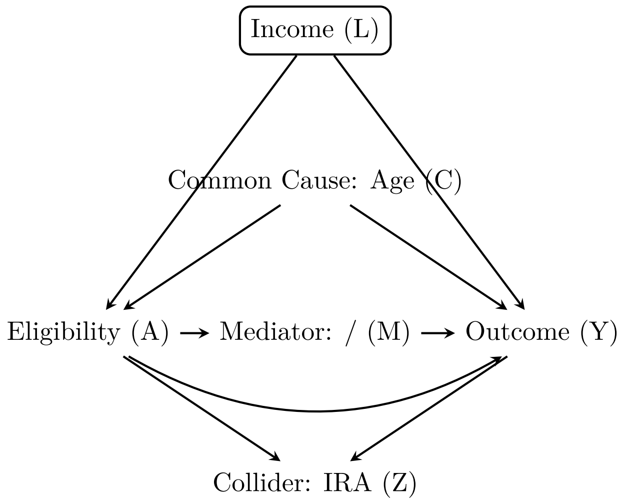

Figure 1: Combined DAG showing collider, mediator, and common cause structures

2) Investigating heterogeneity based on subgroup analysis

Based on the CATE predictions obtained from the causal forest, the data were split into four subgroups sorted on predicted treatment effects. This allows us to assess if the effect differs across individuals who were predicted to benefit more or less from eligibility.

Table 1: ATE estimates by predicted CATE quartile

Quartile	Estimate	SE	CI_low	CI_high
Q1 (lowest predicted CATE)	3729.07	1165.49	1444.72	6013.43
Q2	6236.09	2371.53	1587.90	10884.28
Q3	8009.43	2100.33	3892.79	12126.07
Q4 (highest predicted CATE)	13226.84	3326.52	6706.86	19746.82

We find that the estimated treatment effect is positive in all four quartiles. This implies that, on average, 401(k) eligibility has a positive impact on net financial assets for the complete distribution, even for those benefiting the least from the policy. However, this treatment effect differs across subgroups. The estimated ATE increases across the four predicted-CATE quartiles: approximately $3729 in Q1, $6236 in Q2, $8009 in Q3, and $\$1.3227 \times 10^4$ in Q4. This implies that individuals who were predicted to benefit the most from being eligible based on the causal forest, indeed do experience a larger treatment effect, and thus the causal forest performs rather well.

This also suggests meaningful treatment effect heterogeneity. While adjacent quartiles have overlapping confidence intervals (e.g. Q1 overlaps with Q2), the 95% confidence interval for Q1 [$1444.72, $6013.43] does not overlap with the confidence interval for Q4 [$\$6706.86$, $\$1.974682 \times 10^4$]. This indicates that the treatment effect in the highest quartile is significantly different from the lowest quartile, meaning that these groups differ in their estimated treatment effects. The width of confidence intervals also increases among the quartiles, as there is greater uncertainty among the treatment effect for these benefiting the most.

Overall, this implies that 401(k) eligibility increases net financial assets for all groups, but the estimated gains are substantially larger for the subgroup with the highest predicted treatment effects and there is thus proof of treatment effect heterogeneity. This is because individuals in the highest quartile likely have a higher income, meaning that they can benefit more from becoming eligible than individuals with a lower income.

3) Investigating heterogeneity using the Efficient Influence Function

The Efficient Influence Function (EIF) was used to construct a debiased estimator for assessing treatment effect heterogeneity.

Here, we estimate the variance based on the efficient influence function. The key property that we use is that the expectation of this efficient influence function is equal to 0. Therefore, by taking expectations and isolating the variance, we write the variance of quantities that were already estimated. This includes the estimated CATE, estimated ATE, the propensity score and outcome regression. We then plug these estimates into the formula to calculate $\hat{\sigma}^2$. We can then also compute the efficient influence function values, which we can use to calculate the standard error and corresponding 95% confidence interval.The full derivation is described more formally and in detail in the appendix.

Our estimate for the variance is approximately $\hat{\sigma}^2$ = 8.20e+06, (SE = 3.94e+07), with a 95% confidence interval of [-6.90e+07]. As ^2 denotes the variance, this obviously cannot be negative in reality. However, as we use the debiased estimator, it can be shown that estimates are not constrained to be positive. To see this, consider that the first term of the EIF should always be positive (as it is squared), however if the correction term is larger than the first term, then the found estimate can be negative.

A variance estimate of 0 would indicate a homogenous treatment effect, while an estimate different from 0 indicates a heterogenous treatment effect. Our estimate of the variance is thus positive, which would normally be indicative of a heterogenous treatment effect. However, we can also see that there are very large confidence intervals, which include 0. As this is the case, this implies that we cannot reject the null hypothesis of a homogenous treatment effect, which means we cannot conclude that there is significant heterogeneity in the impact of being eligible for 401(k) schemes on net financial assets for different households.

Because the confidence intervals are rather large; ($\hat{\sigma}^2 = 8.1978899 \times 10^6$), 95% CI [$-6.8996527 \times 10^7$, 8.5392307×10^7]), we also inspect what drives this value. For this purpose, we inspect whether this result is caused by a few observations with extreme EIF values, which we define as values in the top percent of efficient influence function values. We find that the largest EIF value equals 2.1858792×10^{11}, while the 99th percentile equals 8.1554129×10^9, which we consider as extreme.

These extreme EIF values can originate from three 'sources', as can be derived from the formula. First, they may come from large deviations of the estimated CATE from the ATE. Second, they can come from extreme propensity scores. Third, they can result from large outcome residuals. Hence, we compare these for the top 1% vs top 99% of the population.

As shown in Table X, the extreme EIF values are the result of large residuals. The mean absolute residual among the top 1% most influential observations is 2.7122637×10^5, compared with 1.579485×10^4 for the rest of the sample. These extreme values for these observations are the result of higher net financial assets on average: 2.5911691×10^5, compared with 1.559544×10^4. This indicates that several high-asset households have observed outcomes that are substantially larger than predicted by the outcome model. On the other hand, the CATE-ATE deviations are also large, (dit moet nog beter) The propensity scores among these extreme EIF values are rather normal.

4) Estimation of the causal effect using a orthogonal R-learner

The causal forests that were used in the previous phases have the drawback that they learn the causal effect conditional on many variables, making it both difficult to learn with good accuracy and difficult to visualize. In this fourth phase, we therefore estimate the average effect of eligibility for enrollment in a 401(k) plan on net financial assets conditional only on income. For this purpose, an orthogonal R-learner is used, controlling for the same set of covariates as before to ensure valid confounding adjustment. While both DR learner and the R-learner both are theoretically sound to use in this case, it was ultimately decided to implement the R-learner. The R-Learner was chosen because it is more robust to the local overlap failures visible in the propensity score plot. Where $\hat{p}$ is extreme, the DR-Learner amplifies errors through IPW division and requires explicit clipping, introducing bias. The R-Learner instead naturally downweights those observations via $W\_R = (A - \hat{A})^2$, requiring no clipping at all.

We estimated the CATE of 401(k) eligibility on net financial assets using an R-Learner, with nuisance components estimated via random forests and the final stage modeled as a weighted linear regression on income. Compared to the causal forest approach in question 1, 2, 3, which attempts to quantify the value of the heterogeneity, the R-Learner more directly answers the research question by characterizing how the treatment effect varies with income. The linear final stage improves interpretability but adds the assumption that there is a linear relationship between income and the treatment effect, introducing misspecification risk if the true relationship is nonlinear. Next to this, working with the R-learner also assumes that the Neyman orthogonality condition holds, which mean that the conditional means of the residuals are zero. This is a strong assumption, and if it does not hold, the estimates may be biased. The resulting CATEs show a similar pattern to the causal forest: mostly positive values with a right tail of larger effects, though the R-Learner produces a more concentrated distribution. The derived ATE is 8130.6298003 (SE: 51.5309338), consistent across folds, confirming stable calibration. The cross-fold plot shows all five folds intersecting near the mean ATE and following similar linear trends, though high-income observations with large CATEs visibly pull the fit. Diagnostically, the predicted outcomes E[Y] are right-skewed, reflecting heterogeneity in outcome potential and possible sensitivity to outliers. Propensity scores are approximately uniform between 0 and 0.8, confirming the overlap assumption holds and that IPW-based estimation remains stable. Together, these diagnostics support the reliability and robustness of the estimated CATEs and ATE.

Addendum:

Q1:

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

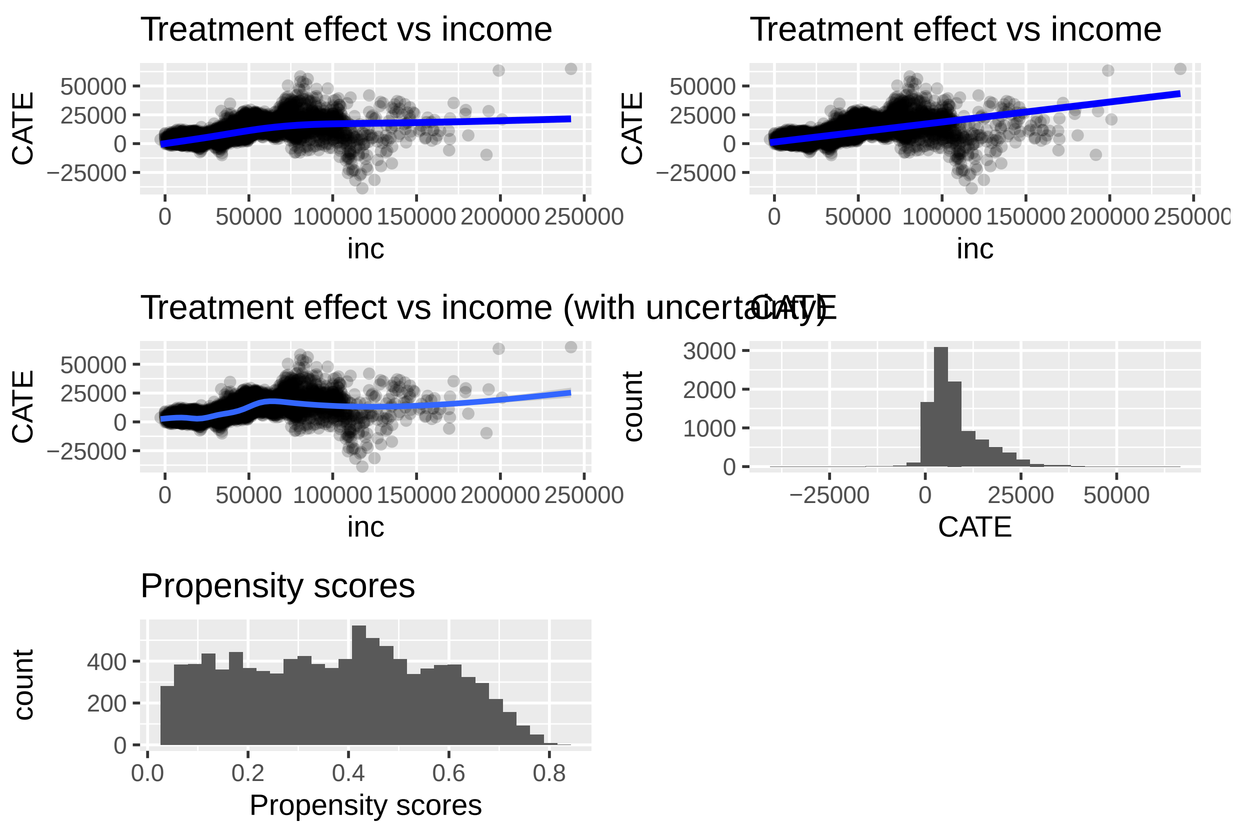

Figure 2: Figures related to Question 1

Q3

Starting from the efficient influence function,

$$
\begin{aligned}
\phi(O) =& \{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}^2 - \sigma^2 \\
& + 2\{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}\left\{\frac{A}{P(A=1 \mid L)} - \frac{1-A}{P(A=0 \mid L)}\right\} \\
& \times \{Y - E(Y \mid A, L)\}.
\end{aligned}
$$

Since the efficient influence function has mean zero,

$$E\{\phi(O)\} = 0.$$

Taking the expectation of the full expression gives

$$
\begin{aligned}
E\Bigg[&\{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}^2 - \sigma^2 \\
&+ 2\{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}\left\{\frac{A}{P(A=1 \mid L)} - \frac{1-A}{P(A=0 \mid L)}\right\} \\
&\times \{Y - E(Y \mid A, L)\}\Bigg] = 0.
\end{aligned}
$$

We can therefore isolate σ^2 as

$$
\begin{aligned}
\sigma^2 = E\Bigg[&\{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}^2 \\
&+ 2\{E(Y^1 - Y^0 \mid L) - E(Y^1 - Y^0)\}\left\{\frac{A}{P(A=1 \mid L)} - \frac{1-A}{P(A=0 \mid L)}\right\} \\
&\times \{Y - E(Y \mid A, L)\}\Bigg].
\end{aligned}
$$

Thus, we estimate σ^2 by replacing the unknown quantities by estimates and taking the empirical average:

$$
\begin{aligned}
\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\Bigg[&\{\widehat{E}(Y^1 - Y^0 \mid L_i) - \widehat{E}(Y^1 - Y^0)\}^2 \\
&+ 2\{\widehat{E}(Y^1 - Y^0 \mid L_i) - \widehat{E}(Y^1 - Y^0)\}\left\{\frac{A_i}{\widehat{P}(A=1 \mid L_i)} - \frac{1-A_i}{\widehat{P}(A=0 \mid L_i)}\right\} \\
&\times \{Y_i - \widehat{E}(Y_i \mid A_i, L_i)\}\Bigg].
\end{aligned}
$$

Here, $\widehat{E}(Y^1 - Y^0 \mid L_i)$ is the estimated CATE from the causal forest, $\widehat{E}(Y^1 - Y^0)$ is the estimated ATE, $\widehat{P}(A = 1 \mid L_i)$ and $\widehat{P}(A = 0 \mid L_i)$ are the estimated propensity scores, and $\widehat{E}(Y_i \mid A_i, L_i)$ is the estimated outcome regression.

The estimated influence-function values are

$$
\begin{aligned}
\hat{\phi}_i =&\{\widehat{E}(Y^1 - Y^0 \mid L_i) - \widehat{E}(Y^1 - Y^0)\}^2 - \hat{\sigma}^2 \\
&+ 2\{\widehat{E}(Y^1 - Y^0 \mid L_i) - \widehat{E}(Y^1 - Y^0)\}\left\{\frac{A_i}{\widehat{P}(A=1 \mid L_i)} - \frac{1-A_i}{\widehat{P}(A=0 \mid L_i)}\right\} \\
&\times \{Y_i - \widehat{E}(Y_i \mid A_i, L_i)\}.
\end{aligned}
$$

The standard error was estimated as

$$
\widehat{SE}(\hat{\sigma}^2) = \frac{sd(\hat{\phi}_i)}{\sqrt{n}},
$$

and the 95% confidence interval was computed as

$$
\hat{\sigma}^2 \pm 1.96 \times \widehat{SE}(\hat{\sigma}^2).
$$

Q4 figures:

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```